



PrimeCell® Infrastructure AMBA™ 3 AXI™ to AMBA 3 AHB™ Bridges (BP137) Revision: r0p1 **Technical Overview**

This technical overview describes the functionality of the AXI to AHB bridges in the following sections:

- *Preliminary material* on page 2
- *About the AXI to AHB bridges* on page 3
- *Functional description* on page 6
- *Physical data* on page 8
- *Signal descriptions* on page 11.

1 Preliminary material

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1.1 Release information

Table 1 lists the changes to this document.

Table 1 Change history

Date	Issue	Confidentiality	Change
17 December 2004	A	Non-Confidential	First issue for r0p0
08 September 2005	B	Non-Confidential	Second issue for r0p1

1.2 Proprietary notice

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1.4 Product status

The information in this document is for a final product, that is a developed product.

1.5 Web address

<http://www.arm.com>

2 About the AXI to AHB bridges

The configurations that you can implement for an AXI to AHB bridge are described in:

- *AXI to ARM11 AHB-Lite bridge*
- *AXI to AHB-Lite master bridge*
- *AXI to AHB-Lite master bridge with OVL assertions* on page 4
- *AXI to AHB-Lite slave bridge* on page 4

2.1 AXI to ARM11 AHB-Lite bridge

The AXI to ARM11 AHB-Lite bridge implements an AXI slave port and an ARM11 AHB-Lite master port to enable, for example, an ARM11 AHB-Lite system to be connected to an AXI master.

The ARM11 extensions to the standard AHB specification are defined in the *Systems IP ARM11 AMBA AHB Extensions Specification*.

For more information on AHB-Lite see *AHB Lite Overview* (ARM DVI 0044).

2.2 AXI to AHB-Lite master bridge

The AXI to AHB-Lite master bridge that Figure 1 shows implements the AXI to ARM11 AHB-Lite bridge together with the components required to provide an AHB-Lite master interface. This enables, for example, an AHB-Lite system to be connected to an AXI master.

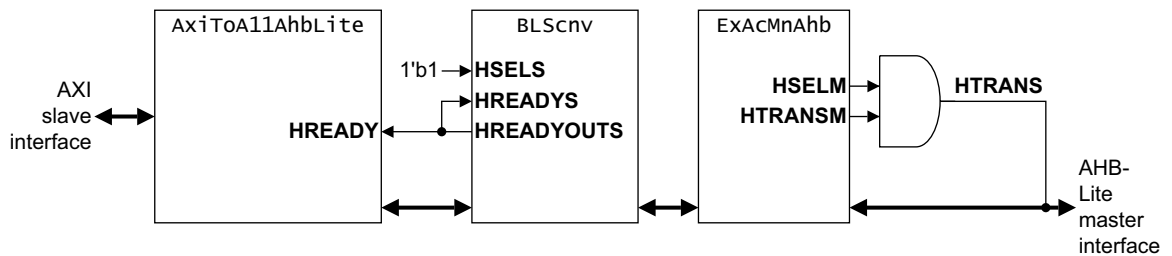


Figure 1 AXI to AHB-Lite master bridge

The other components in Figure 1 are:

Byte lane strobe converter

The byte lane strobe converter, BLScnv, is an AHB slave gasket that supports the byte lane strobe functionality defined in the *Systems IP ARM11 AMBA AHB Extensions Specification*.

Exclusive access monitor

The exclusive access monitor, ExAcMn, is an AHB slave gasket that supports the exclusive access functionality defined in the *Systems IP ARM11 AMBA AHB Extensions Specification*. It enables up to two masters exclusive access to a slave.

2.3 AXI to AHB-Lite master bridge with OVL assertions

If you do not require the complexity of the configuration described in *AXI to AHB-Lite master bridge* on page 3 then the component can be configured as a bridge that converts the ARM11 AHB-Lite master interface into an AHB-Lite master interface. This is achieved by tying-off the following signals:

- **HPROT[5:4]** are not propagated. There is no support for exclusive, cache allocate transfer indication.
- **HBSTRB** and **HUNALIGN** are not propagated. There is no support for unaligned or sparse transfers.
- **HRESP[2]** is tied LOW. There is no support for exclusive response.

The AXI to AHB-Lite bridge wraps the AxiToA11Lite bridge selectively to create an AHB-Lite master interface. The bridge contains OVL assertions that guard against unsupported features of the AXI protocol. Figure 2 shows the bridge.

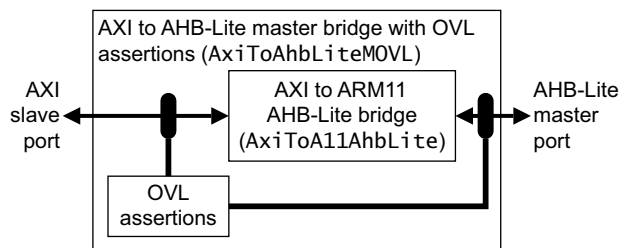


Figure 2 AXI to AHB-Lite master bridge with OVL assertions

2.4 AXI to AHB-Lite slave bridge

The AXI to AHB-Lite slave bridge that Figure 3 on page 5 shows implements the AXI to ARM11 AHB-Lite bridge together with the components required to provide an AHB-Lite slave-gasket interface. This enables, for example, an AHB-Lite slave to be connected to an AXI system.

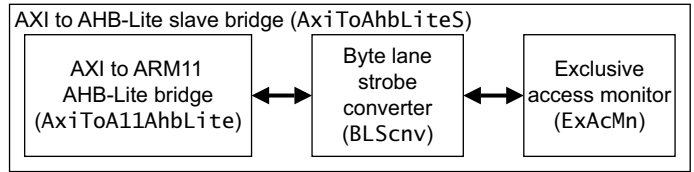


Figure 3 AXI to AHB-Lite slave bridge

The bridge uses the byte lane strobe converter and exclusive access monitor as described in *AXI to AHB-Lite master bridge* on page 3.

3 Functional description

The AXI to ARM11 AHB-Lite bridge has the following features:

- It translates AXI transactions to AHB-Lite bursts with the following restrictions:
 - exclusive accesses are available by using the bridge and an external exclusive access monitor
 - unaligned and sparse accesses are available by using the AXI ARM11 AHB-Lite bridge together with an ARM11-capable AHB interconnect, or an external byte lane strobe converter.
- The bridge only supports one outstanding transaction at any one time on each address channel. This is described in *Interface attributes* on page 7.
- Initiation of AHB write bursts are related to AXI address and write channel transfers:
 - there is a minimum latency of two clock cycles from the acceptance of an AXI transaction, **AWVALID** or **ARVALID** asserted, to the start of the AHB transaction when **HTRANS** is driven to **NONSEQ**.
 - the first data transfer in a write burst waits for the AXI address, **AWVALID**, and the first AXI write transfer, **WVALID**
 - subsequent data transfers in a write burst wait for the AXI write transfer, **WVALID**, and busy cycles are issued on the AHB domain until the data arrives.
 - the AXI write response is issued on the completion of the final write data transfer of the burst.
- Initiation of AHB read bursts are related to AXI address and read channel transfers:
 - the first AHB address is issued on or after the arrival of the AXI address transfer, **ARVALID**
 - subsequent AHB transfers wait for the acceptance of the read data by the AXI read channel, **RREADY**, and busy cycles are issued on the AHB domain until the data has been accepted.
- Only the **OKAY**, **EXOKAY** and **SLVERR** responses are generated:
 - an error response returned on any data transfer of an AHB write burst causes a write response of **SLVERR**
 - read responses are translated on a per transfer basis

- an AHB XFAIL response results in an AXI OKAY response. An AHB OKAY response to an exclusive access results in an AXI EXOKAY response.
- All AXI transactions are mapped, where possible, to an equivalent AHB burst:
 - wrapping and incrementing bursts of length 4, 8, and 16 are converted into the equivalent length AHB bursts
 - incrementing bursts of length 1, wrapping bursts of length 2, and all fixed address transactions are converted to sequences of AHB SINGLE bursts
 - any burst that crosses a 1KB address boundary is converted into INCR bursts with the burst restarted, **HTRANS** is overridden to **NONSEQ**, when the boundary is crossed
 - all remaining incrementing and wrapping bursts are converted into AHB INCR bursts
- The address transfer ID is used for read and write transfers:
 - **ARID** is used for the associated **RID**, and for **HMASTER** during AHB read transfers
 - **AWID** is used for the associated **BID**, and for **HMASTER** during AHB write transfers.
- HDL code supplied as Verilog.

3.1 Interface attributes

The slave interface attributes for the AXI to AHB bridges are described in Table 2.

Table 2 Slave interface attributes

Attribute	Description	Value
Write acceptance capability	The maximum number of active write transactions that a slave can accept.	1
Read acceptance capability	The maximum number of active read transactions that a slave can accept.	1
Write interleave depth	The number of active write transactions for which the slave can receive data. This is counted from the earliest transaction.	1
Read data reorder depth	The number of active read transactions for which a slave can transmit data. This is counted from the earliest transaction.	1

4 Physical data

Physical data is provided in:

- *AC characteristics*
- *Gate count* on page 10.

4.1 AC characteristics

Timing closure might be difficult to obtain in a complex system because the AXI to AHB gasket contains combinatorial paths. The easiest way to rectify this is to use an AXI register slice or buffer on any channels that do not meet timing. This, however, incurs a latency penalty that might not be desirable if interfacing to an AHB or APB interrupt controller.

Trial synthesis using Synopsys Design Compiler was performed with a clock cycle of 5ns, 200MHz, skew 2.5%, on the Artisan SAGE HS library for the TSMC CL013G process, sample 0.13µm cell library, slow-slow process corner, with no resulting timing violations.

The timing constraints are provided in:

- *AXI to ARM11 AHB-Lite bridge timing constraints*
- *AXI to AHB-Lite master bridge timing constraints* on page 9
- *AXI to AHB-Lite slave bridge timing constraints* on page 9.

AXI to ARM11 AHB-Lite bridge timing constraints

Figure 4 on page 9 shows the constraint strategy used for the AXI ARM11 AHB-Lite bridge core during development. The figures denote the percentage of clock cycle permitted for each function. The timing constraints are:

- paths from inputs to registers are permitted 30% of the clock cycle
- paths from registers to outputs are permitted 20% of the clock cycle
- combinatorial through paths between AXI and AHB ports are permitted 30% of the clock cycle.

Note

These figures are also relevant for the AXI to AHB-Lite master bridge with OVL assertions.

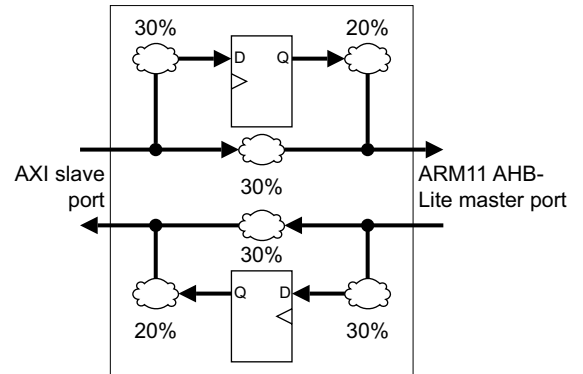


Figure 4 AXI to ARM11 AHB-Lite bridge timing constraints

AXI to AHB-Lite master bridge timing constraints

The timing constraints are:

- paths from inputs to registers are permitted 40% of the clock cycle
- paths from registers to outputs are permitted 30% of the clock cycle
- combinatorial through paths between AXI and AHB ports are permitted 40% of the clock cycle.

AXI to AHB-Lite slave bridge timing constraints

The timing constraints are:

- paths from inputs to registers are permitted 40% of the clock cycle
- paths from registers to outputs are permitted 30% of the clock cycle
- combinatorial through paths between AXI and AHB ports are permitted 40% of the clock cycle.

4.2 Gate count

Table 3 lists the estimated gate count in the library specified in *AC characteristics* on page 8.

Table 3 Gate counts

Bridge	NAND2X1 equivalents	
	32-bit data width	64-bit data width
AXI to ARM11 AHB-Lite	3700	4800
AXI to AHB-Lite master	6700	8300
AXI to AHB-Lite master with OVL assertions	3500	4500
AXI to AHB-Lite slave	6800	8300

———— **Note** ————

These gate count estimates do not include scan logic.

The ID_WIDTH parameter was set to 4 for all estimates.

5 Signal descriptions

The signals that the bridges use are described in:

- *AXI to ARM11 AHB-Lite bridge*
- *AXI to AHB-Lite master bridge* on page 12
- *AXI to AHB-Lite master bridge with OVL assertions* on page 13
- *AXI to AHB-Lite slave bridge* on page 13.

Note

The upper value of some bus widths is provided as a name to indicate that the number of signal lines in the bus is derived from user-defined generics or parameters. These are described in the *PrimeCell Infrastructure AMBA 3 AXI to AMBA 3 AHB Bridges (BP137) Design Manual*.

5.1 AXI to ARM11 AHB-Lite bridge

The AXI to ARM11 AHB-Lite bridge signals that Figure 5 on page 12 shows are standard AXI signals as described in the *AMBA AXI Protocol Specification* and ARM11 AHB-Lite, signals as described in the *Systems IP ARM11 AMBA AHB Extensions Specification* and the *AMBA Specification (Rev 2.0)*.

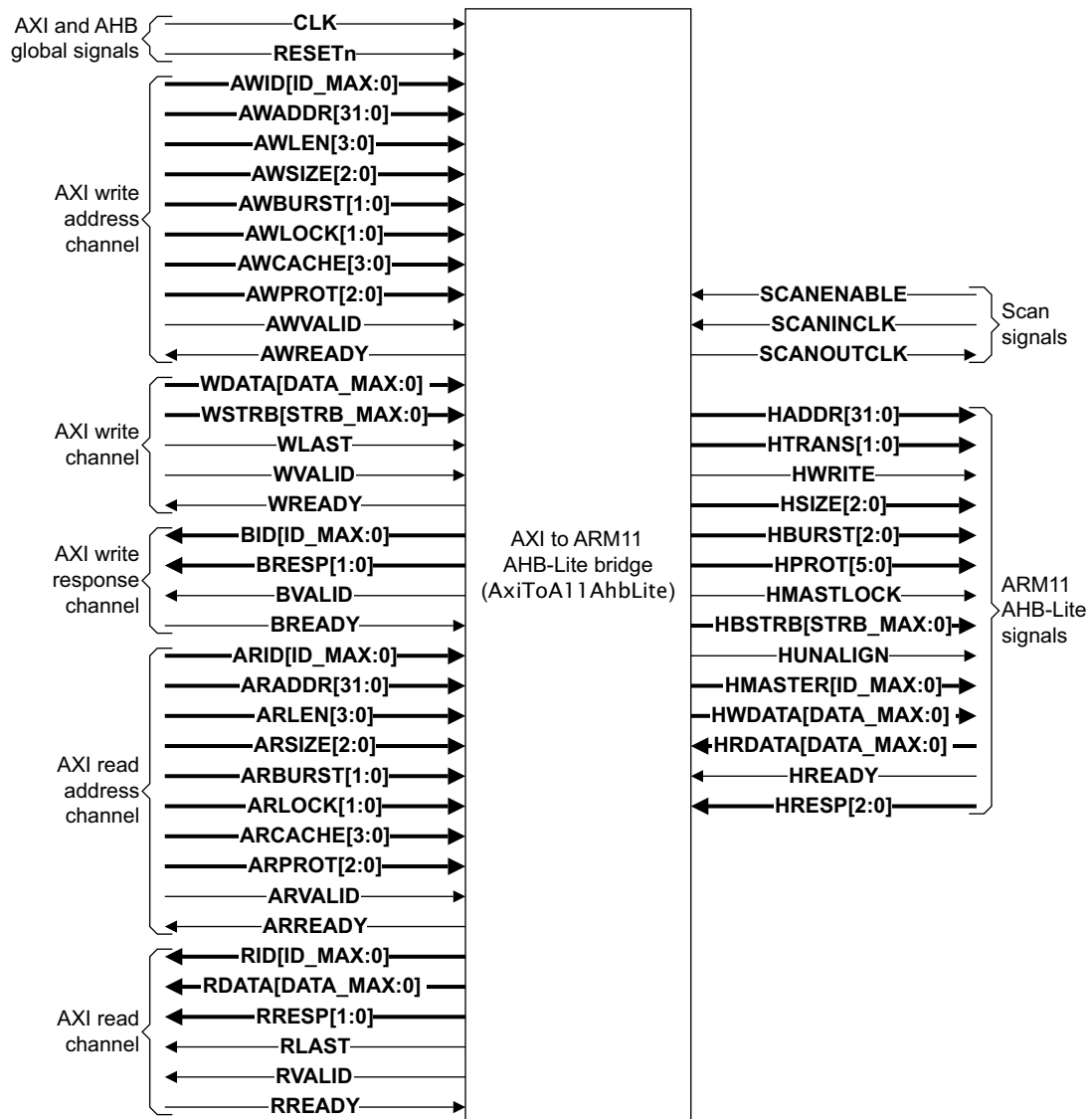


Figure 5 AXI to ARM11 AHB-Lite bridge signal connections

5.2 AXI to AHB-Lite master bridge

The AXI to AHB-Lite master bridge signals are standard:

- Figure 5 shows the AXI and AHB global clock and reset signals

- Figure 5 on page 12 shows the scan signals
- the *AMBA AXI Protocol Specification* describes the AXI signals and Figure 5 on page 12 shows them
- the *AMBA Specification (Rev 2.0)* describes the AHB-Lite signals and Figure 6 shows them.

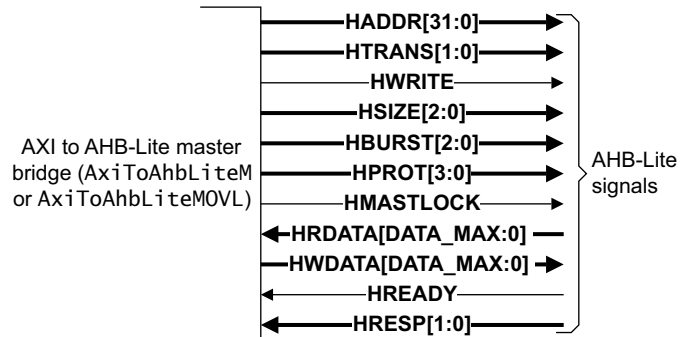


Figure 6 AXI to AHB-Lite master bridge signal connections

5.3 AXI to AHB-Lite master bridge with OVL assertions

The AXI to AHB-Lite master bridge signals are standard:

- Figure 5 on page 12 shows the AXI and AHB global clock and reset signals
- Figure 5 on page 12 shows the scan signals
- the *AMBA AXI Protocol Specification* describes the AXI signals and Figure 5 on page 12 shows them
- the *AMBA Specification (Rev 2.0)* describes the AHB-Lite signals and Figure 6 shows them.

5.4 AXI to AHB-Lite slave bridge

The AXI to AHB-Lite slave bridge signals are standard:

- Figure 5 on page 12 shows the AXI and AHB global clock and reset signals
- Figure 5 on page 12 shows the scan signals
- the *AMBA AXI Protocol Specification* describes the AXI signals and Figure 5 on page 12 shows them

- the *AMBA Specification (Rev 2.0)* describes the AHB-Lite signals and Figure 7 shows them.

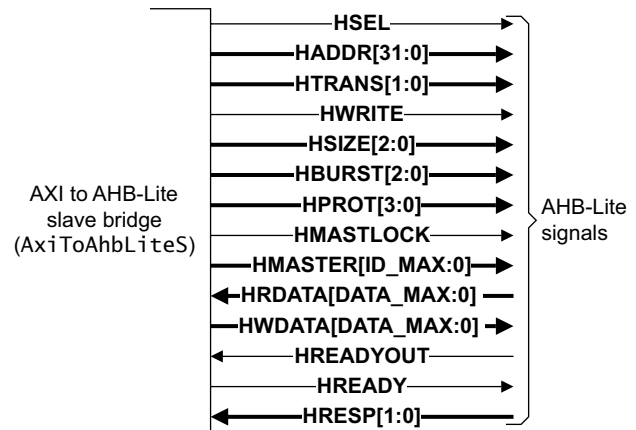


Figure 7 AXI to AHB-Lite slave bridge signal connections